Muhammad Zain

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Profile

Final-year Software Engineering student (graduating April 2027) seeking full-time roles starting May 2027. Builds, ships, & operates LLM-backed web apps end to end, including a live React & Python product rated 99 on Lighthouse.

Education

University of Calgary — Calgary, AB
Bachelor of Science (Software Engineering) — *September 2022 - April 2027*

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, C++

AI & LLM: Gemini API, LLM Integration, Prompt Engineering, Structured JSON Output, Model Evaluation

Web & Graphics: React, Three.js, WebGL, Vite, Flask, REST APIs, JWT/OAuth, Node.js (Familiar)

Cloud & Tooling: Docker, GitHub Actions (CI/CD), Git, GCP (Cloud Run), Cypress, JUnit, Agile, AWS (Familiar)

Experience

Enactus Canada — Calgary, AB
Software Developer Intern — *September 2025 - May 2026*

- Designed a spatiotemporal data pipeline in sprints, aligning ERA5 reanalysis, satellite vegetation indices, storm events, & static terrain onto a canonical 30 km grid to produce weekly wildfire risk scores across Alberta.
- Enforced leakage-safe temporal validation & nearest-neighbor grid joins through peer code review, eliminating coordinate drift, record duplication, & forward-looking time contamination across multi-year experiments.
- Surfaced 40-60% of fires within the top 1% of ranked cells under 500:1 class imbalance, tuning gradient-boosted models for PR-AUC & Recall@Top-K & adding an Isolation Forest anomaly score agreed in design review.
- Standardized reproducible outputs including Parquet tables, serialized model artifacts, & interactive risk-map visualizations, presenting findings to technical & non-technical stakeholders for operational handoff.

Projects

Resume & Career Coach (RCC) — *January 2026 - June 2026*
Python, Flask, React, PostgreSQL, Docker, Gemini LLM

- Engineered anti-hallucination guardrails via structured JSON prompts, curated skill coverage, & SHA-256 caching.
- Shipped resumecoach.app, a live AI web app scoring resume-to-job fit via a hybrid token-overlap & LLM pipeline.
- Built an LLM-powered mock interview engine with adaptive questioning, persisted sessions, & browser voice I/O.
- Eliminated false scores on unreadable uploads with layout-aware parsing & an extractability guard returning 422.

Lumen Pendulum — *October 2025 - November 2025*
TypeScript, Three.js, WebGL, Vite, GitHub Actions

- Delivered a public interactive 3D WebGL visualization holding 60fps under continuous animation & drag control.
- Removed frame stutter from garbage collection with a zero-allocation render loop over pre-allocated GPU buffers.
- Implemented real-time physics from Lagrangian mechanics with a 4th-order Runge-Kutta integrator at 8 sub-steps.
- Enabled safe iteration with a typed module architecture separating physics, rendering, & input concerns.

Software Testing & QA — *January 2026 - April 2026*
Java, JUnit, Mockito, Cypress, TypeScript, PIT, JaCoCo

- Raised instruction coverage from 14.5% to 100% by creating & executing 173 unit tests across two libraries.
- Exposed injected faults hidden behind 99% line coverage via PIT mutation testing, reaching an 89% mutation score.
- Authored the full end-to-end test layer for a 6-person team's web product across 56 automated browser tests.
- Drove 10 defects to resolution through a full Jira lifecycle, peer-reviewing reproduction steps & severity.

ProfileHub Performance Evaluation — *January 2026 - April 2026*
Python, Flask, React, MySQL, GCP, Locust, SciPy

- Reached 0% request failures across 72 load runs by fixing a connection-pool leak that drained slots under load.
- Deployed a containerized Cloud Run service with managed SQL, signed-URL object storage, & a 4-worker runtime.
- Identified non-linear saturation past 15 concurrent users, a 38% latency jump that set the team's capacity ceiling.
- Measured 2.4x higher throughput on read-heavy vs. write-heavy workloads (5.12 vs 2.14 requests per second).